NextMove - Architectural Model

View 1 of 3: logical architecture - layers, Django apps, data stores and external services

PRESENTATION

Student browser

Django templates styled with Bootstrap 5 (loaded from a CDN)
Plain JavaScript chat: sends with fetch, polls every 3 seconds
Unread badge, scam warning banner, Report this listing link

Staff browser

Django admin site (staff accounts only)
Dashboard cards on the admin home page
Moderation, report and user actions

HTTPS requests

HTTPS requests

APPLICATION
Django 6.1
Python 3.14

Request pipeline

Security, Session, Common, CSRF protection, Authentication, Messages and X-Frame-Options middleware, then URL routing to the views

accounts

Custom User (UoN email only)
Signup, email OTP, login
Forgot password by code
Profile, change password

marketplace

Category, Listing, ListingImage
Browse, post, edit, My listings
Favorites, soft delete
Keyword hold on save

messaging

Conversation, Message
Inbox, chat, JSON polling
scam.py: 8 rule groups
Unread count, read receipts

reports

Report model and form
One open report per student and listing
Dashboard statistics tag
Admin report actions

Django admin (staff only)

Users: verify, disable, enable (no add, no delete) | Listings: approve, reject, hold (no hard delete)
Reports: mark reviewed, dismiss, reject (removes the listing) | Conversations: message counts only - staff never read chat text

ORM queries

photo files

OTP and reset emails

DATA AND SERVICES

MySQL database nextmove_prod

10 application tables (accounts_user, listings,
images, keywords, flags, favorites,
conversations, messages, reports)
plus Django sessions and admin log

Local media folder

media/listings/YYYY/MM/
JPEG, PNG, WebP, GIF
up to 5 MB, at most 6 per listing
served by Django in development

Gmail SMTP (external)

smtp.gmail.com port 587 with TLS
sends OTP and password-reset emails
sender and app password come from .env

CONFIG

Configuration

.env (DB_PASSWORD, GMAIL_ADDRESS, GMAIL_APP_PASSWORD, MAX_IMAGES, MAX_IMAGE_BYTES, ALLOWED_FORMATS) read with python-dotenv; config/settings.py

Holds one of the four differentiating features (email OTP, keyword hold, live chat, scam alerts)

External or temporary service

Data store

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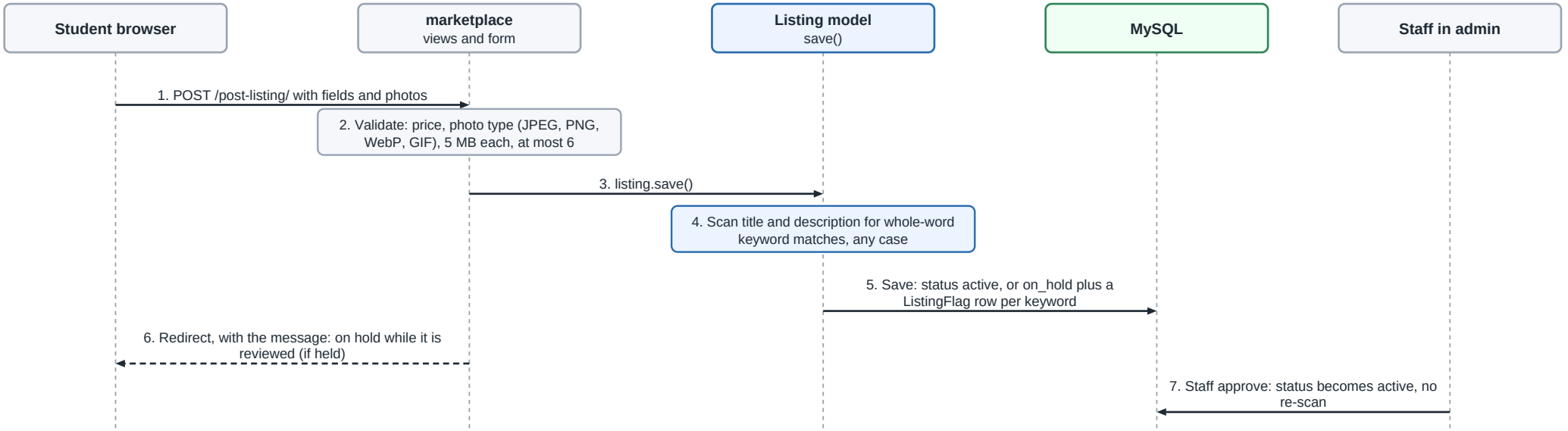
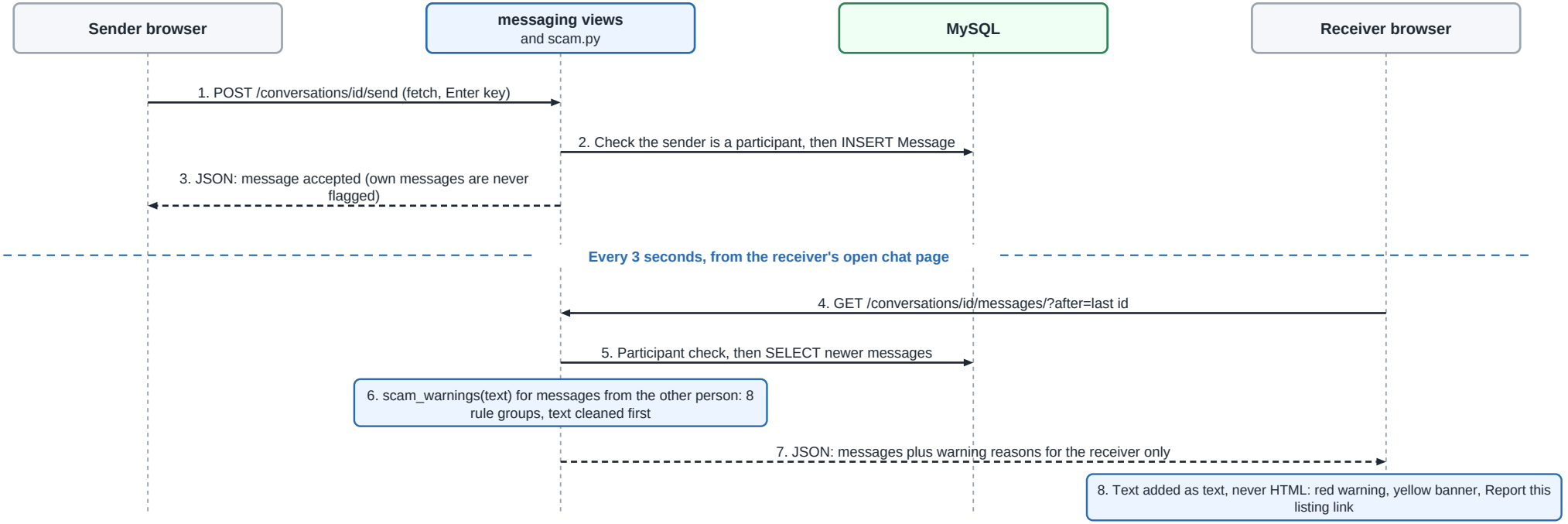
View 2 of 3: key runtime flows - keyword hold on posting a listing, and live chat with scam alerts

A. Posting a listing: validation, keyword scan and hold



B. Live chat with scam alerts (nothing about the alerts is stored)



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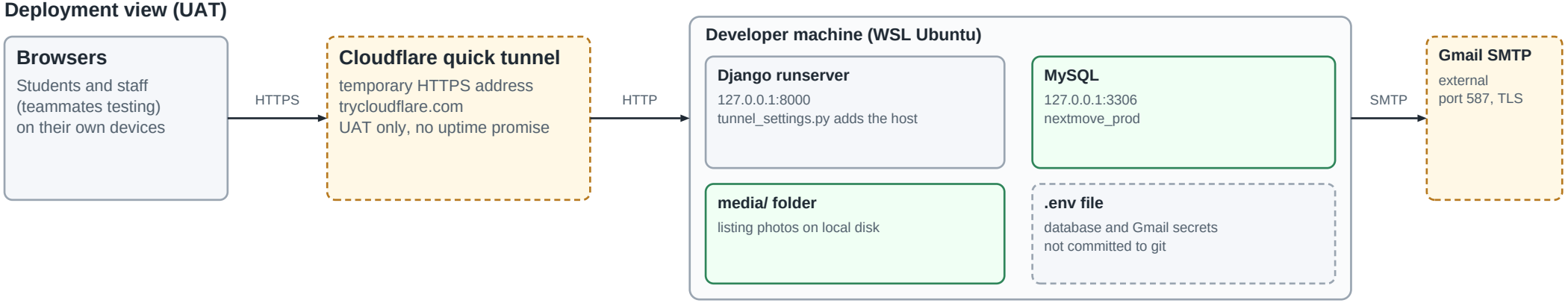
View 3 of 3: deployment used for user acceptance testing, and the main architectural decisions

Deployment view (UAT)



Local development skips the tunnel: the browser talks straight to 127.0.0.1:8000. For real use the team recommends a small server, a production web server, HTTPS, debug off, MySQL and protected media (future work, Section 14 of the report).

Architectural decisions and their trade-offs

Decision	Why	Trade-off
One Django monolith with four apps: accounts, marketplace, messaging, reports	Built-in authentication, forms, CSRF protection and admin keep the code small and easy to test	Less front-end interactivity than a single-page application
Chat polls every 3 seconds with plain JavaScript	Works with the development server and needs no extra services	Messages can take up to 3 seconds to appear; websockets are the next step
Keyword scan runs inside Listing.save()	Every path (post, edit, reopening a sold listing) is covered in one place; it runs only on new or changed text, so a staff approval is not undone	Whole-word matching still holds some honest listings, for example Weed killer
Scam checks run when messages are served, and nothing is stored	Chats stay private from staff and no database change is needed	Warnings are recalculated each time and leave no audit trail
Staff work in the Django admin with custom actions	Permission-protected and quick to build; chat text is never shown to staff	Looks less polished than bespoke admin pages
Soft delete: listing status removed	The listing disappears for students (404) but staff keep the record for reports	No restore action yet
MySQL database and local media storage	Matches the unit environment and needs no cloud account	Photos sit on one disk and are reachable by URL, so not production-grade
Email OTP through Gmail SMTP, secrets in .env	Proves control of a UoN mailbox and also powers password reset	Depends on Gmail limits; the OTP is stored unhashed for its 15-minute life